

EXPERIMENTAL DATA MANAGEMENT SYSTEM DOCUMENTATION

Milestone 1 and Milestone 2

1. Problem Definition

This project implements an Experimental Data Management System that models how experimental data is collected, stored and evaluated.

The system is inspired by real world testing environments such as Kenya Bureau of Standards, where multiple trials are conducted and measurements are validated against predefined standards.

2. System Design

The system is designed using object-oriented programming principles:

Classes and their responsibilities

- Experiment
 - ✓ Represents a complete experiment
 - ✓ Contains multiple trials
 - ✓ Evaluates overall experiment result
- Trial
 - ✓ Represents a single experimental run
 - ✓ Contains multiple measurements
 - ✓ Determines pass/fail outcome
- Measurement
 - ✓ Base class representing a measured variable
 - ✓ Stores variable name and value

3. Milestone 1 Implementation

The concepts below were implemented

- Classes and objects
- Attributes (name, value, trial_id)
- Methods (add_measurement, display)
- Encapsulation of data within objects
- Basic system structure (Experiment → Trial → Measurement)

The system at this stage was capable of:

- Storing experimental data
 - Displaying structured results
- ### 4. Milestone 2 Implementation (Interaction and Control logic)

The system was extended to include:

- a) Object Interaction
 - Experiments contain trials

- Trials contain measurements
- Objects interact through method calls

b) Inheritance

A base Measurement class was extended into specialised classes:

- TemperatureMeasurement
- PHMeasurement
- WeightMeasurement

This promotes code reuse

c) Polymorphism

Each measurement type implements:

Is_valid() in which different classes use the same method name but differently.

- d) Control Logic
 - Each trial evaluates whether measurements meet standards
 - Each experiment aggregates trial results

Decision rules:

- Trial = FAIL if any measurement is invalid
 - Experiment = FAIL if any trials fail
5. System Workflow

Experiment → Trials → Measurements → Validation → Decision

What changed:

In milestone 1, the system focused on basic object modelling with classes such as experiment, trial, and measurement.

In milestone 2, the system was extended to include interaction, inheritance and decision-making logic.

What failed:

Initially the system did not support validation of measurements, meaning all data was accepted regardless of correctness

Why it failed:

This happened because the initial design focused only on data structure and not on behaviour or decision-making processes.

How it was fixed:

The issue was resolved by introducing validation methods in measurement classes and implementing control logic in the trial and experiment classes to evaluate results.

6. Statistical Relevance

The system aligns with statistical principles:

- Trials represent replications
- Measurements represent observed variables
- Validation represents quality control processes

7. Limitations

- No file storage yet
- No statistical analysis yet
- No exception handling yet

These will be addressed in future milestones

8. Conclusions

The system successfully models a real-world experimental environment using object-oriented programming. It demonstrates foundational concepts and introduces decision making logic necessary for managing experimental data.